CSE2525 · DATA MINING · 2026–27

Introduction to Data Mining

What data mining is, how this course is run, and why you should tread with caution — this year with an information-access focus.

A. Anand (head lecturer) · with J. Urbano & N. Tömen · lecture 1 study deck

PART I

What is data mining?

From a cholera map in 1854 to malware graphs today — and where it sits next to machine learning.

QUESTION

London, 1854: hundreds are dying of cholera and no one knows why. All you have is a list of the **addresses of the dead**. How would you find the cause?

No germ theory yet, no microscope data — just locations on a map.

Think — then → for the answer.

LONDON, 1854

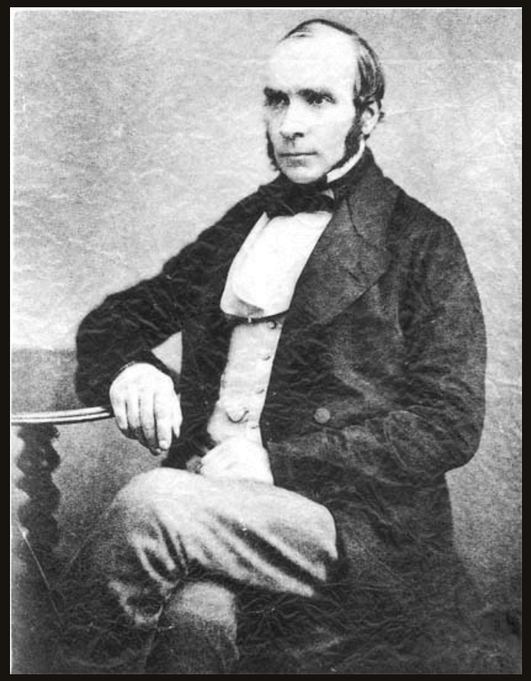
The first data miner — John Snow

THE PROBLEM

A cholera outbreak tore through Soho — **~600 people dead in days**. The accepted theory was *"bad air"* (miasma). Nobody knew the real cause.

— Snow, a physician, suspected the water — but had no proof.

— All he had was a list of addresses of the dead. How do you find a cause in that?

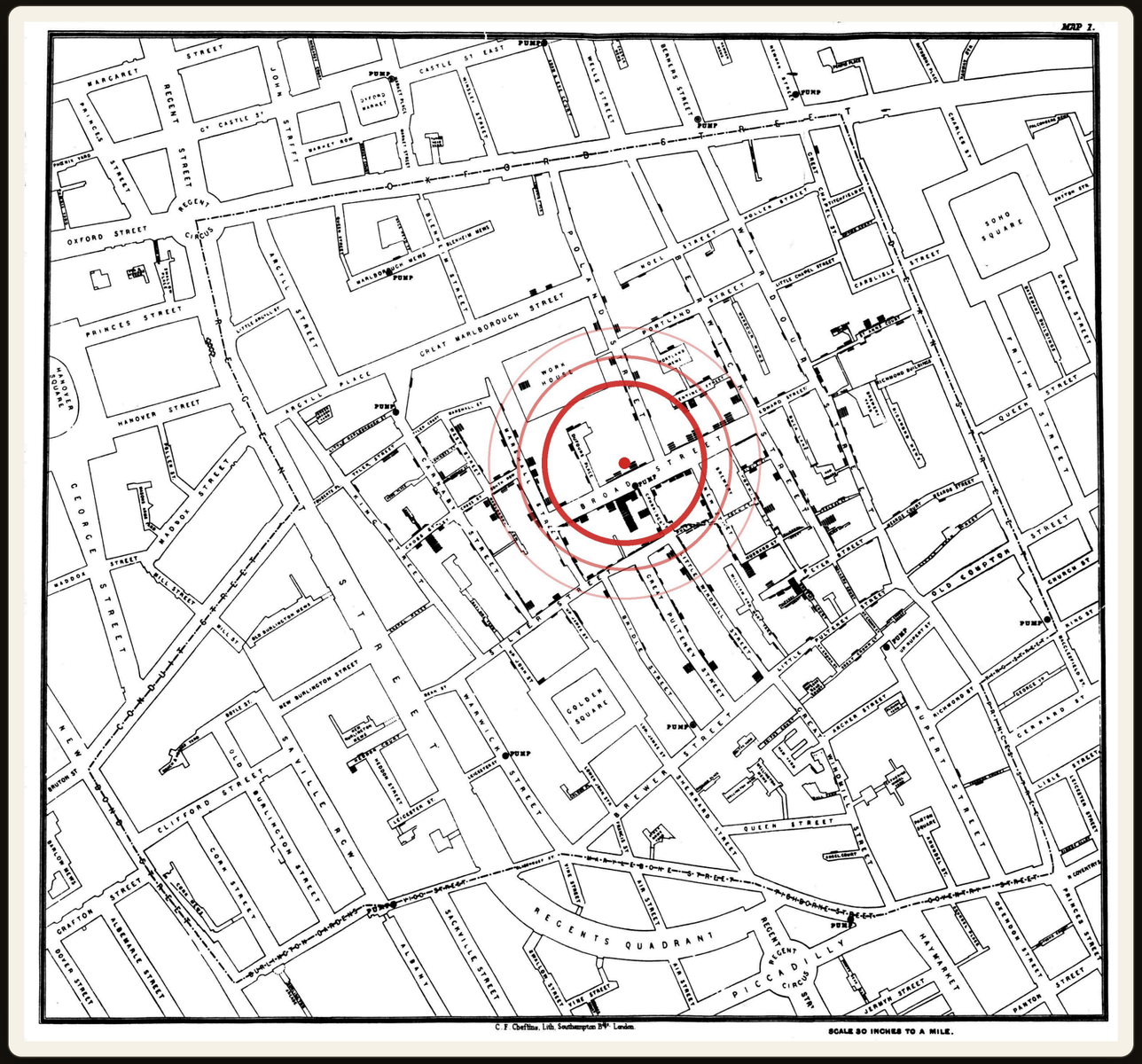


HE PLOTTED EVERY DEATH ON A MAP



Each red dot is a cholera death. Snow marked them street by street — and a dense cluster began to emerge in the middle of Soho, while whole blocks nearby stayed clear.

ONE PUMP EXPLAINED IT ALL



The deaths ringed one water pump on Broad Street. Snow had its handle removed — and the outbreak stopped. The cluster was the evidence: data mining before the name existed.

ALEXANDRIA, 3RD CENTURY BC

Another origin story — the first retrieval system

~500,000 scrolls
AN UNUSABLE PILE

→

The Pinakes catalogue
AUTHOR · SUBJECT · SHELF

→

Any scroll, found
INFORMATION RETRIEVAL

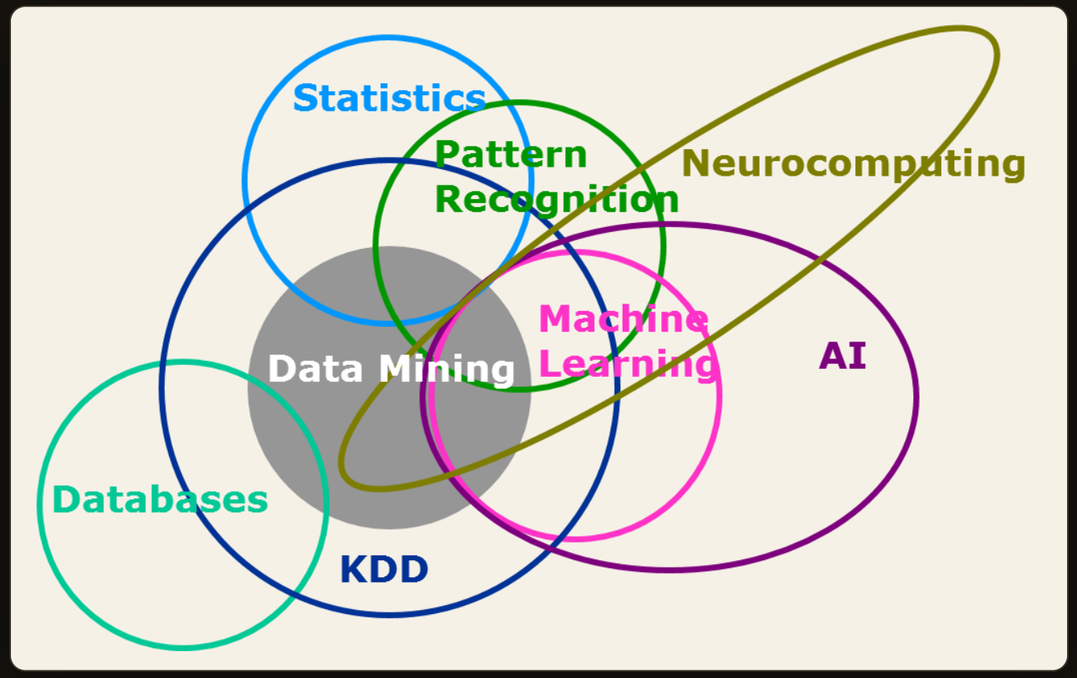
SAME IDEA, A DIFFERENT QUESTION

Callimachus catalogued the Library of Alexandria — the Pinakes, arguably the first information-retrieval system: structure the data so any work can be *found*.

Snow found a pattern in the data; the Pinakes found the needle. Both are data mining — turning a pile of data into something that answers a question. This year's focus: information access.

BLOGS.SAS.COM

Data mining sits at an intersection



It borrows from statistics, databases, machine learning, pattern recognition and AI — the overlap at the heart of Knowledge Discovery in Databases (KDD).

MANY PHRASINGS, ONE IDEA

What *is* data mining?

WORKING DEFINITION

The primary goal of data mining is to **extract useful information from a large volume of data** and transform it into an **understandable structure** for further use.

— "…searching and analyzing a large batch of raw data to identify patterns and extract useful information."

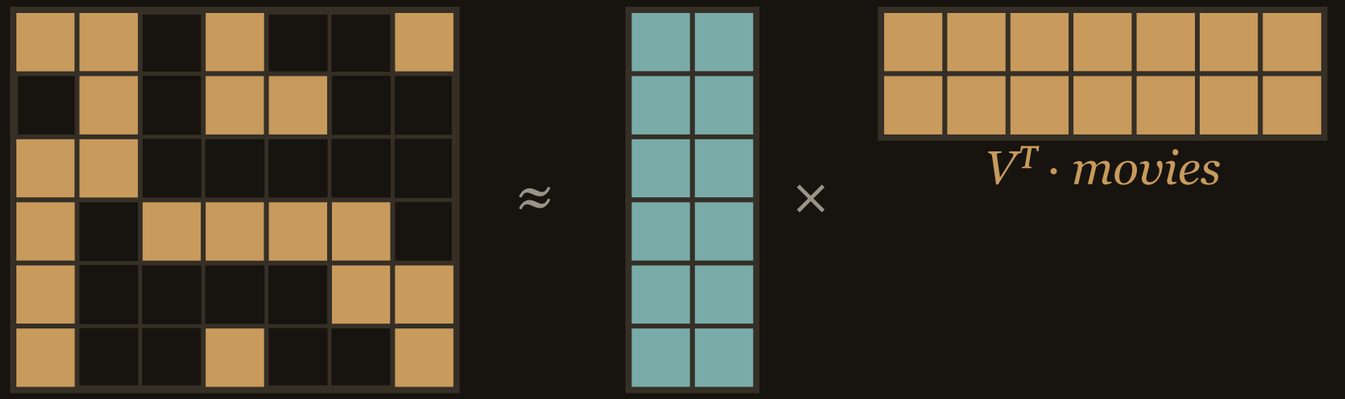
— "…the development of models for data in order to extract information from it."

— "…analyzing data from different perspectives and summarizing it into useful information."

— And crucially — it is done by humans.

EXAMPLE 1 OF 3 · RECOMMENDATION

Netflix: what should you watch next?



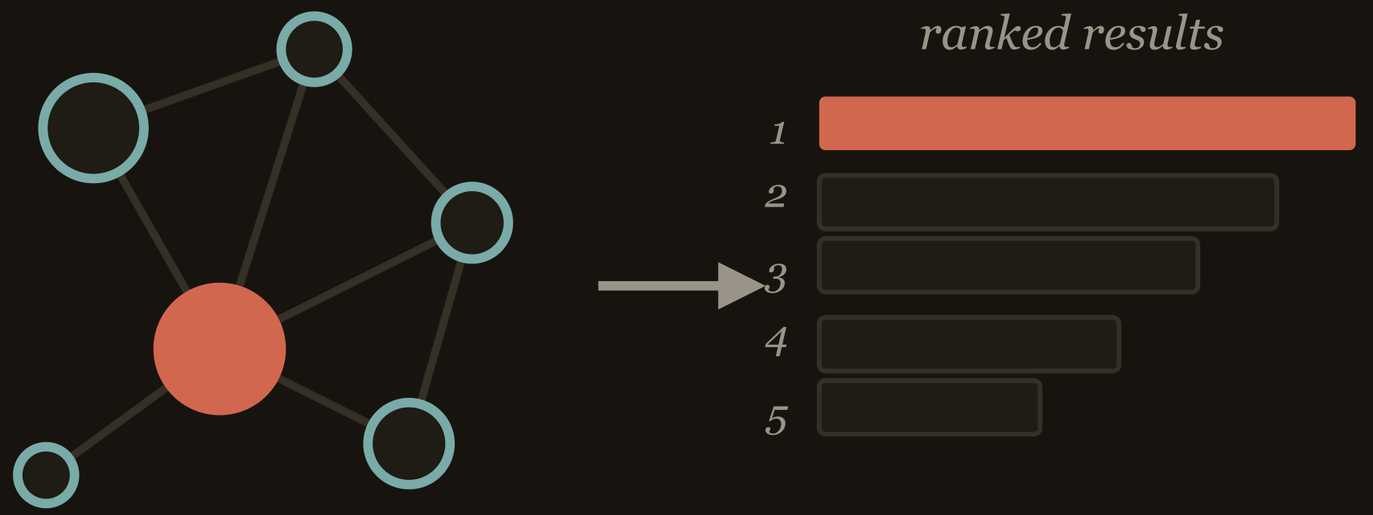
R · ratings (mostly ?) *U · users*

similar users like similar films

Distances / similarity | Embeddings | Matrix factorization | Clustering | Dimensionality reduction (PCA)

EXAMPLE 2 OF 3 · WEB SEARCH

Google: find & rank the whole web

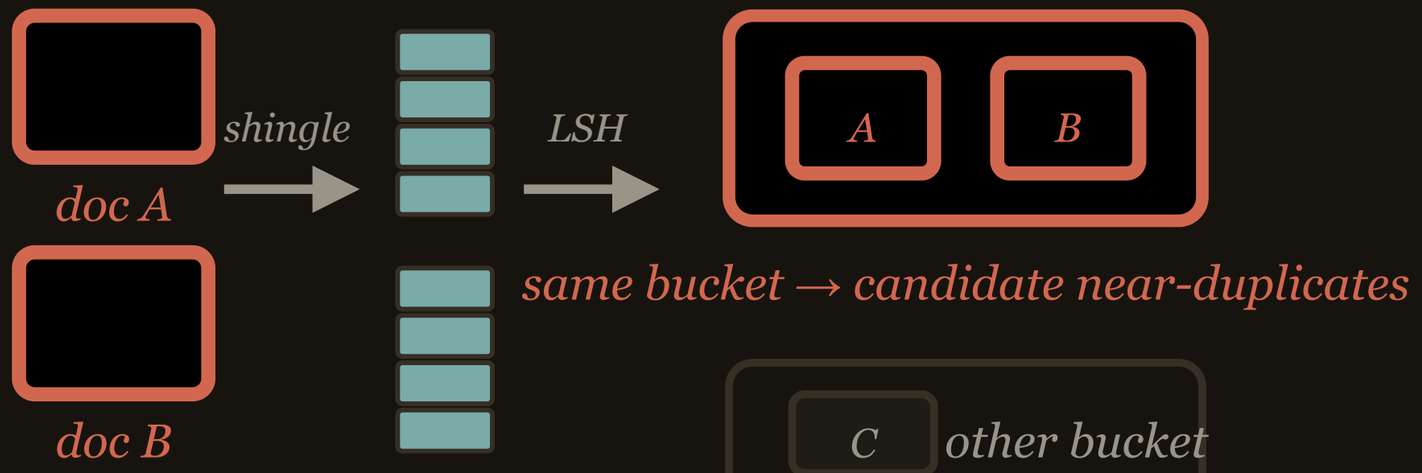


as a link graph — node size = PageRank

Inverted index | Graph mining · PageRank | Graph embeddings | Ranking | Retrieval | Evaluation

EXAMPLE 3 OF 3 · AT SCALE

Deduping the web: near-copies among billions



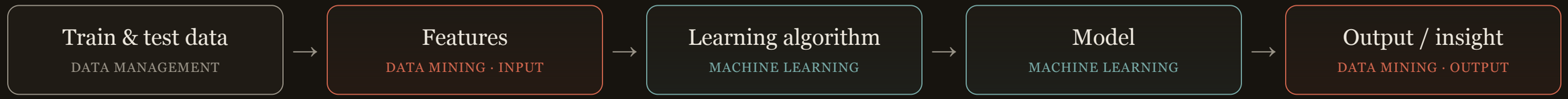
MinHash signatures

Jaccard distance · Shingling · MinHashing · LSH / hashing

Fast-filter → slow-check

SAME PIPELINE, THREE DIFFERENT JOBS

Data mining vs. machine learning vs. data management



Data management — store & serve the data **Machine learning** — the blue core: algorithm & model **Data mining** — the orange ends: input & output

One shared pipeline — what differs is which stage each field owns.

RECAP

Where data mining lives

THE DIVISION OF LABOUR

Data management stores and serves the train/test data. Machine learning owns the blue middle — which algorithm, how to train it, and the resulting model.

Data mining owns the orange ends: choosing the input features from domain knowledge, and turning the output into insight. Often the learning here is unsupervised.

A HEALTHCARE EXAMPLE

Mining first, then predicting

DATA MINING PHASE

Analyse a large database of patient records — symptoms, diagnostics, treatments, outcomes — with **clustering**, **association rules**, **pattern discovery** to *uncover patterns* (e.g. fever + cough + shortness of breath → a specific respiratory illness).

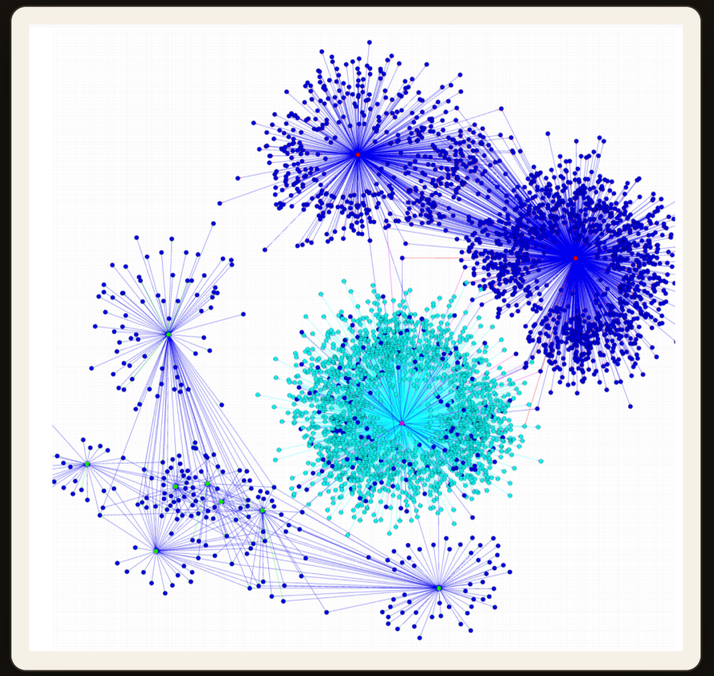
MACHINE LEARNING PHASE

Use those discovered patterns to **train a model**. A new patient's symptoms → the model predicts likely diagnoses and suggests treatments from historical success rates.

Data mining finds the structure; machine learning predicts with it.

CYBER SECURITY · MALWARE FAMILIES

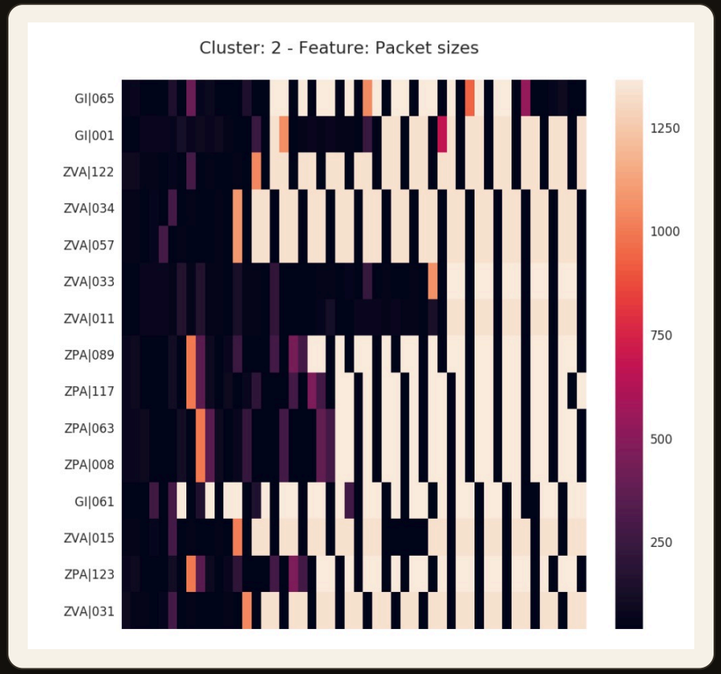
A real example: network / graph data



Connections between hosts form a huge graph. Clustering it reveals behavioural profiles of malware families — no labels needed (Nadeem et al., 2020).

MAKING THE RESULT INSIGHTFUL

Visualise groups with heatmaps



Row = connection · column = time · cell colour = bytes transferred. A cluster becomes a readable behavioural signature.

THE WHOLE LOOP

These are all data-mining skills

THE DATA-MINING LOOP

Take the data as is, but select features using domain knowledge → cluster with clustering methods → make the gained knowledge insightful (e.g. heatmaps).

Feature choice, the right algorithm, and a human-readable result — that end-to-end judgement is the craft.

THE THREE V'S

Why is data mining hard?

Volume

- Too big to fit / scan naively
- Need **scalable** algorithms

Velocity

- Data arrives fast / streams
- Need **approximate yet accurate** estimates

Variety

- Many types & distributions
- Must **adapt** methods to each

These three pressures are exactly why we need hashing, indexing, embeddings and dimensionality reduction.

PART II

The course & logistics

The team, what's new this year, and how everything runs — goals, schedule, labs, grading, prerequisites.

WHO TEACHES THIS COURSE

Three instructors, three verticals

Avishek Anand · head lecturer

- Assoc. prof., Web Information Systems
- IR, NLP, Explainable AI
- Teaches: **distances, embeddings, graphs** — text & graph mining

Julián Urbano

- Asst. prof., Computer Science (TU Delft)
- Music IR, retrieval evaluation, significance testing
- Teaches: **retrieval, rankings, evaluation**

Nergis Tömen

- Asst. prof., Intelligent Systems
- Bio-inspired vision, neuromorphic computing
- Teaches: **matrices, PCA, decompositions, recommenders**

WHAT'S DIFFERENT THIS YEAR

This year: an information-access focus

THE SHIFT

The course is reoriented around information access — how we find, rank, and evaluate the right information inside massive data, not just describe it.

Content swap vs. last year: out go anomaly detection and sketching; in come retrieval, ranking, and evaluation.

— One lab assignment this year — down from two.

— No Kaggle competition this year.

— New reading: Introduction to Information Retrieval (Manning, Raghavan & Schütze) for selected chapters.

WHAT YOU SHOULD WALK AWAY WITH

Course goals

THE PROMISE

You learn Data Mining at three levels: the **algorithms**, the **core concepts** behind them, and the **practical skill** of choosing and applying them.

- Key algorithms — implement them; know their strengths, weaknesses, and *when* to use them.
- Core concepts — including the theory, and which concepts fit which data types.
- Practical skills — pick the right method for a problem and make it work on real data.

THE SPINE OF THE WHOLE COURSE

Three verticals: Distances · Matrices · Retrieval

Distances

- Similarity & metrics, DTW
- Text & graph embeddings
- Clustering
- Hashing & MinHash / LSH (counting lives here)

Matrices

- Representation, properties, operations
- Factorization & decompositions
- Dimensionality reduction (PCA)

Retrieval

- Indexing & the inverted index
- Ranking
- Evaluation
- Near-duplicate / similar-item search

This year's shift: counting folds into Distances, and Retrieval becomes its own vertical — the information-access focus. Every topic hangs off one of these three.

MONDAYS 13:45 · THURSDAYS 10:45 · EEMCS AMPERE · LABS TUE 08:45

Schedule

Week	Monday 13:45	Thursday 10:45
9 Nov	Intro	Distances
16 Nov	Retrieval	Matrices & PCA
23 Nov	Rankings	Evaluation
30 Nov	Embeddings	Matrix decomposition
7 Dec	no lecture	Recommender systems
14 Dec	Hashing	Sketching
21 Dec–1 Jan	❄ Christmas break — no teaching	
4 Jan	Invited lecture	Indexing
11 Jan	Graphs	Bias & explanations
18 Jan	Exam preparation	no lecture
25 Jan	📝 **Data Mining EXAM · Mon 25 Jan 13:30**	

⚑ The first lab is in Week 5 (Tue 8 Dec) — there are no lab sessions until Week 5.

17 LECTURES · 1 LAB

How the teaching works

- 17 lectures — 13 content, 2 invited, 1 intro, plus Q&A sessions.
- In the room we do case studies — how do you actually apply what you learned to a real problem?
- Complete schedule lives on Brightspace.

WHERE THE CONTENT COMES FROM

Course material

ON BRIGHTSPACE (REQUIRED)

Lecture slides (after each class), lab exercises (start of each week), and reading materials — selected book chapters and papers.

- Free via the TU Delft digital library: *Mining of Massive Datasets* · *Data Mining*.
- New this year — *Introduction to Information Retrieval* (Manning, Raghavan & Schütze) for the retrieval / ranking / evaluation chapters. Free online: nlp.stanford.edu/IR-book
- Only the selected chapters uploaded to Brightspace are examinable.

MANDATORY · IN STUDENT PAIRS

Lab sessions

- One lab assignment this year, in student pairs — Tuesday mornings (08:45), assistance and feedback on the spot.
- Two parts: algorithm implementation (Distances / Matrices / Retrieval) and a pipeline (transform → analyse → visualise). No Kaggle this year.
- Example: implement DTW & PCA from scratch, then build and evaluate the pipeline.
- Everything is built from scratch — no scikit-learn or other ML toolkits.
- Help channels: Queue, Mattermost, Answers EWI (tag CSE2525), Weblab, Peer. Please do not use e-mail — it won't be answered. Pair up early and register on Brightspace.

LABS 30% · EXAM 70%

Lab evaluation & grading

HOW THE LAB IS GRADED — 30% OF THE FINAL GRADE

Algorithmic component — evaluated automatically. Pipeline component — peer-reviewed (each submission gets 4 reviews; a penalty if you skip your own reviews — we spot-check quality).

No Kaggle competition this year. No posted solutions — ask for help during the lab sessions.

THE ASSESSMENT

Final exam

— Final exam = 70%, on WebLab (Osiris + WebLab registration). Open and multiple-choice questions.

— No programming questions this year · closed book, calculator allowed · one resit.

— No partial transfer from previous years. mytimetable.tudelft.nl is authoritative for dates.

WHAT WE ASSUME YOU ALREADY KNOW

Expected prior knowledge

— Discrete maths — sets, intersections, unions.

— Linear algebra — matrix multiplication, projections, eigenanalysis.

— Probability & statistics — Gaussians, correlation, covariance.

— Graph theory — adjacency matrix, degree, clique, bipartite graph, shortest path.

— Data structures — hash tables and indexes. Programming — Python.

— Machine learning — logistic regression, random forest, SVM, …

Based on CSE1100/1305/1200/1205/1210 and CSE2510/2520. Not enforced — you are responsible for your own study success.

RECAP OF PART I

Logistics in one slide

KEY POINTS

Questions through online tools and at sessions. One mandatory lab — automated tests + peer review (no Kaggle). Closed-book exam.

This year's focus: information access — retrieval, ranking and evaluation replace anomaly detection and sketching.

Feedback only: dm-cs-ewi@tudelft.nl.

PART III

Course content

Everything, organised under the three verticals.

DISTANCES · MATRICES · RETRIEVAL

The whole course on one map

Distances

- Similarity, metrics, DTW
- Text & graph embeddings
- Clustering
- Hashing · MinHash / LSH

Matrices

- Representation, properties, operations
- Factorization & decompositions
- Dimensionality reduction (PCA)

Retrieval

- Indexing · inverted index
- Ranking
- Evaluation
- Similar-item / near-duplicate search

When a new topic appears, ask: *is this fundamentally a distance, a matrix, or a retrieval problem?*

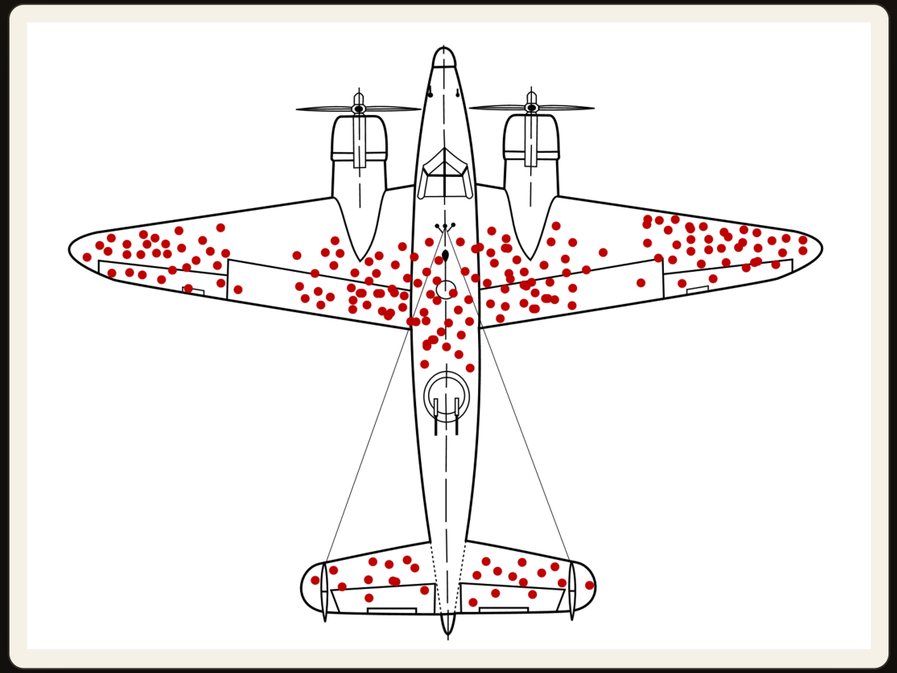
PART IV

A word of caution

Don’t fool yourself. The rest of this lecture is cautionary tales about results that look great and are wrong.

WWII · ABRAHAM WALD

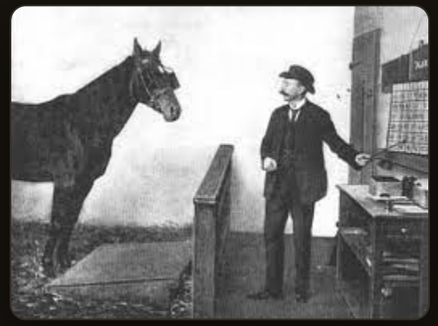
Survivorship bias



Red dots = damage on planes that returned. Where should you add armour? Not where the dots are — where they aren't (engines, cockpit): planes hit there never came back. Your data is only the survivors.

THE HORSE THAT COULD “COUNT”

The Clever Hans effect



Hans seemed to do arithmetic — but was really reading his trainer’s unconscious cues. Models do this too: right answer, wrong reason.

TOO GOOD TO BE TRUE

Shortcuts in learning — Wikipedia citations

THE TASK

Automatically predict which Wikipedia sentences **need a citation**. A neural network scored **100% on the test set** — while humans agreed only **<55%** of the time.

WHAT SHOULD YOU ASK?

100% accuracy on a task humans find genuinely hard is too good to be true — a red flag, not a trophy.

The catch: sentences needing a citation already carried the marker "[ ]". The model learned *"if the bracket is there, predict citation-needed"* — a shortcut, not understanding.

THE CANONICAL ML PARABLE

Shortcut learning — the “Russian tanks”

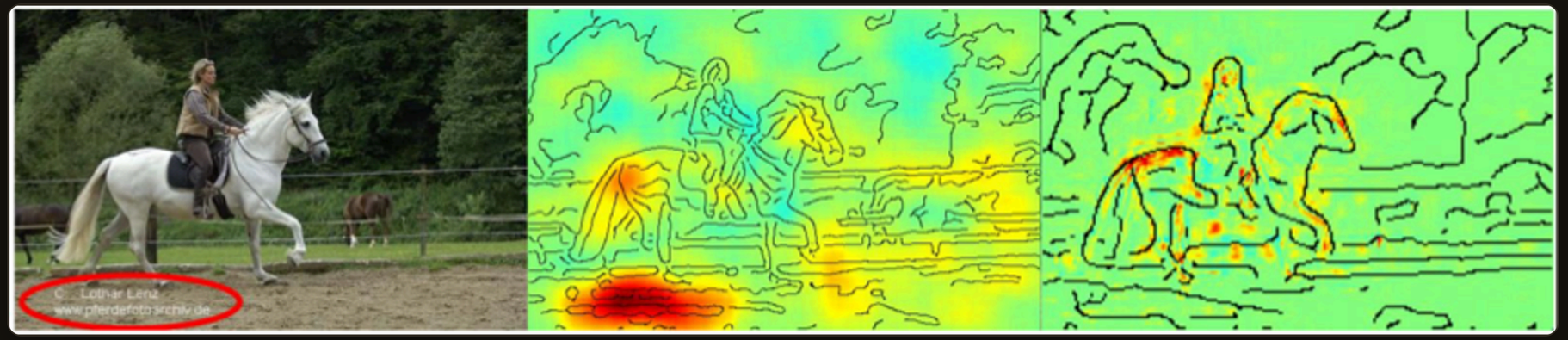




100% accurate in training, worse than random on new data. The network had learned: “if the sky is blue, no tank; otherwise, tank.” It classified the weather, not the tanks.

EXPLAINABLE AI REVEALS THE TRICK

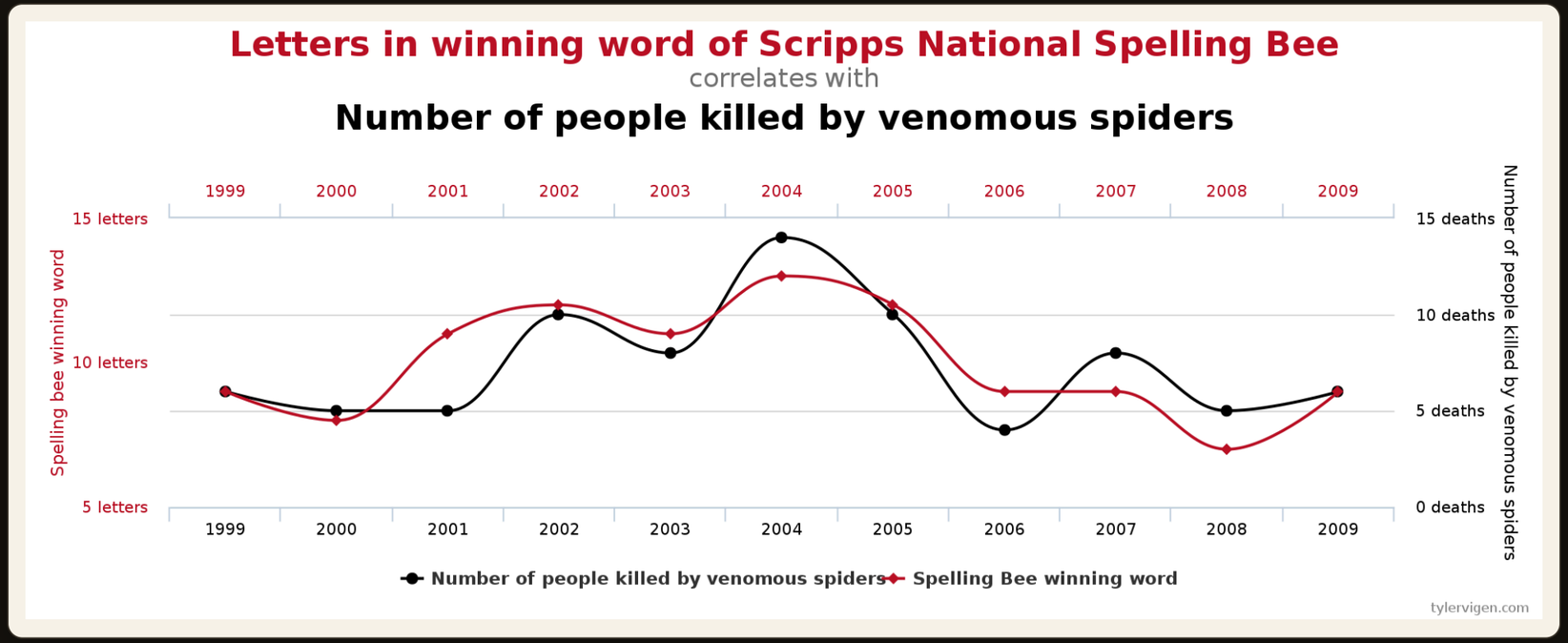
Shortcuts in image models



A classifier labels this “horse” with high confidence. Saliency maps show *why*: it fires on the copyright watermark in the corner. If “© …” → predict horse.

MULTIPLE COMPARISONS

You can always find what you're looking for



Test enough hypotheses and some will "correlate" by pure chance. The multiple-comparisons problem is the cause of many data-mining errors — significance you went fishing for isn't significance.

LECTURE 1 · WRAP-UP

Summary

TAKE AWAY

Data mining = extract useful information from a large volume of data and turn it into an understandable structure. It focuses on the input and output ends of the ML pipeline, often unsupervised.

Organised around three verticals — Distances · Matrices · Retrieval. Grading: labs 30%, exam 70%.

It's called the skill of the 21st century — but tread with caution. Survivorship bias, Clever Hans, shortcut learning and multiple comparisons are waiting to fool you.